

Prompt Engineering

Date: _____

- Technique to create best possible inputs for LLMs.
- Acts as UI to AI reasoning

a) Zero-Shot Prompting → Asking LLMs to do a task without any examples in prompt.
works well for simple, well-known tasks.
• Limitation → Hallucinations in Zero-Shot.

b) Few-Shot Prompting → Gives LLM 2-5 examples of task inside prompt before asking real questions.
Best for classification, extraction, structured outputs.
• Examples are called exemplars which act as mini training set inside prompt.

c) Chain-of-thought (CoT) → Hallucinations increase RISK
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• expand token length
• LLM fills gap when unsure
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• longer step-by-step reasoning creates more chances of errors.

d) Role & System Prompting
• System Prompt → Sets overall rules & behaviour (how AI should always act)
• Role Prompt → Like job description
• What AI should do in specific tasks.

e) Multi-Context Prompting (MCP)
• Divide prompt into layered context - each layer gives a different type of information.
Example:- Context A (Domain Knowledge)
Context B (Constraints)
Context C (Output style)
• Reduces hallucinations.

Prompts are auto created by algorithms, ^{Date:}

f) AutoPrompt (Automatic Prompting)

- Not always human readable.
- Algorithms search for best words, tokens.

g) ReAct Prompting (thinking + Acting in loop)

- Prevents ~~Reasoning~~ Hallucinations



- Pure reasoning (like CoT)
- Invert tool calls for facts

h) Least to Most Prompting

- Model solves hard problem by starting with easiest parts first, then build full solution.

i) Self-Ask Prompting

- i) Technique where model is instructed to generate its own sub-questions first before answering main question.

- ii). deeper reasoning + avoid skipping important steps + useful for causal analysis, reasoning explanation.

j) Meta Prompting (describe how to create other prompts)

prompt about prompting

k) Symbolic Reasoning Prompting

forces model to use formal, logical, symbolic methods instead of natural language guessing.

- Convert statements to formal symbols

Natural Language + Reasoning + Programming Power. ^{Date:}

l) Program-Aided Language (PAL)

- LLM writes short python code snippets to solve tasks instead of doing all reasoning in natural language.

m) Iterative Prompting (ask + review + refine + repeat)

- Refine output step by step through multiple rounds

n) Sequential Prompting (chain multiple prompts one after another)

- Output of one prompt becomes input of next prompt.

Research → Summarize → analyze → present.

- LangChain + CrewAI + LangGraph.

o) Self-Consistency Prompting

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Improvement of CoT.

- Model generates multiple independent reasoning chains.

Final answer chosen by majority

SCP - Analogy (majority is trusted)

TOT - Analogy (team works + kick weak point out + follow best path)

p) Generated Knowledge Prompting

- Model firstly generates relevant background facts. or extra knowledge, then solve the task.

→ Handle Hallucinations in Prompting

- 1) Ask for sources
- 2) Iterative Prompting
- 3) Self-Consistency used
- 4) Combine RAG